

Noise Contrastive Estimation

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In prediction problems, we’re supposed to predict $y \in \mathcal{Y}$ from $x \in \mathcal{X}$. We do this by assuming a joint population distribution $\mathbf{pop}_{XY}$ from which we can sample correct pairs (x, y) and learning a score function $s^\theta(x, y) \in \mathbb{R}$ parameterized by θ such that it assigns a high score to a correct pair and a low score to an incorrect pair. To estimate such a score function, we often use the hinge loss (Appendix A) or the cross-entropy loss (Appendix B)

In **noise contrastive estimation (NCE)**, we choose a “noise” distribution q_Y over $\mathcal{Y}$ and the size of a sample set N and consider the task of distinguishing true samples from fake samples. It underlies many successful methods such as word2vec [7], the generative adversarial networks (GANs) [3], and contrastive predictive coding [8]. It has two popular formulations. 1. **Global**: Infer which of the N samples is true. 2. **Local**: For each individual sample infer if it’s true.

Information theory enables a simple and insightful analysis of NCE. Given any distribution p , if q^θ is a distribution over the same variables parameterized by θ , q^θ is equal to p iff it is the minimizer of the cross entropy between p and q^θ

$$\theta^* \in \arg \min_{\theta} \mathbf{E}_{z \sim p} [-\log q^\theta(z)] \iff q^{\theta^*}(z) = p(z) \quad \forall z$$

assuming the **universality** of q^θ : that is, it is expressive enough to model p so that $p = q^\theta$ for some θ . While universality should be assumed with a grain of salt (e.g., it might require an exponentially large parameter space), it seems to hold in practice with neural networks and greatly simplifies analysis.

1 Global NCE

1.1 Model

The global NCE objective assumes a joint distribution

$$\mathbf{pop}_{I|XY^N}^{q_Y}(i, x, y_1 \dots y_N) := \frac{1}{N} \mathbf{pop}_{XY}(x, y_i) \prod_{j \neq i} q_Y(y_j)$$

That is, we first draw an index $i \in \{1 \dots N\}$ *uniformly* at random and for $j = 1 \dots N$ draw $(x, y_j) \sim \mathbf{pop}_{XY}$ if $j = i$ but otherwise draw $y_j \sim q_Y$. This yields a conditional distribution over N indices

$$\mathbf{pop}_{I|XY^N}^{q_Y}(i|x, y_1 \dots y_N) = \frac{\mathbf{pop}_{Y|X}(y_i|x) \prod_{j \neq i} q_Y(y_j)}{\sum_{k=1}^N \mathbf{pop}_{Y|X}(y_k|x) \prod_{j \neq k} q_Y(y_j)} = \frac{\frac{\mathbf{pop}_{Y|X}(y_i|x)}{q_Y(y_i)}}{\sum_{k=1}^N \frac{\mathbf{pop}_{Y|X}(y_k|x)}{q_Y(y_k)}} \quad (1)$$

Let $H^{q_Y}(I|XY^N)$ denote the conditional entropy of $\mathbf{pop}_{I|XY^N}^{q_Y}$. The following observation is made in [8].

Lemma 1.1. Let $q_Y = \mathbf{pop}_Y$. Then $H^{\mathbf{pop}_Y}(I|XY^N) \geq \log N - I(X, Y)$ where $I(X, Y)$ is the mutual information between $(x, y) \sim \mathbf{pop}_{XY}$.

Proof. By (1),

$$\begin{aligned} & \mathbf{E}_{(i,x,y_1 \dots y_N) \sim \mathbf{pop}_{I|XY^N}^{\mathbf{pop}_Y}} \left[-\log \mathbf{pop}_{I|XY^N}^{\mathbf{pop}_Y}(i|x, y_1 \dots y_N) \right] \\ &= - \underbrace{\mathbf{E}_{(x,y) \sim \mathbf{pop}_{XY}} \left[\frac{\mathbf{pop}_{Y|X}(y|x)}{\mathbf{pop}_Y(y)} \right]}_{I(X,Y)} + \underbrace{\mathbf{E}_{(i,x,y_1 \dots y_N) \sim \mathbf{pop}_{I|XY^N}^{\mathbf{pop}_Y}} \left[\log \sum_{k=1}^N \frac{\mathbf{pop}_{Y|X}(y_k|x)}{\mathbf{pop}_Y(y_k)} \right]}_{\geq \log N} \end{aligned}$$

We will not prove the claim that the second term is at least $\log N$, but it is intuitive since $\mathbf{pop}_{Y|X}(y|x) \approx \mathbf{pop}_Y(y)$ if $y \sim \mathbf{pop}_Y$ and $\mathbf{pop}_{Y|X}(y|x) \gtrsim \mathbf{pop}_Y(y)$ if $y \sim \mathbf{pop}_{Y|X}(\cdot|x)$. A formal proof can be found in [9]. $\square$

Corollary 1.2. $B := \log N - H^{\mathbf{pop}_Y}(I|XY^N) \leq \min \{I(X, Y), \log N\}$.

Proof. The claim that $B \leq I(X, Y)$ follows by rearranging terms in Lemma 1.1. The claim that $B \leq \log N$ follows from the fact that $H^{\mathbf{pop}_Y}(I|XY^N) \geq 0$ (Shannon entropy is nonnegative). $\square$

1.2 Estimation

We use a score function $s^\theta(x, y)$ through the softmax function to estimate $\mathbf{pop}_{I|XY^N}^{q_Y}$

$$p_{I|XY^N}^\theta(i|x, y_1 \dots y_N) := \frac{\exp(s^\theta(x, y_i))}{\sum_{j=1}^N \exp(s^\theta(x, y_j))} \quad \forall i \in \{1 \dots N\}$$

We train the model by minimizing the cross (conditional) entropy between $\mathbf{pop}_{I|XY^N}^{q_Y}$ and $p_{I|XY^N}^\theta$:

$$\bar{H}_\theta^{q_Y}(I|XY^N) := \mathbf{E}_{(i,x,y_1 \dots y_N) \sim \mathbf{pop}_{I|XY^N}^{q_Y}} \left[-\log p_{I|XY^N}^\theta(i|x, y_1 \dots y_N) \right]$$

Note that $\bar{H}_\theta^{q_Y}(I|XY^N) \geq H^{q_Y}(I|XY^N)$ for all θ by the usual property of cross entropy. If $q_Y = \mathbf{pop}_Y$, Corollary 1.2 implies that

$$\begin{aligned} B(\theta) &:= \log N - \bar{H}_\theta^{\mathbf{pop}_Y}(I|XY^N) = \mathbf{E}_{(i,x,y_1 \dots y_N) \sim \mathbf{pop}_{I|XY^N}^{\mathbf{pop}_Y}} \left[\log \frac{\exp(s^\theta(x, y_i))}{\frac{1}{N} \sum_{j=1}^N \exp(s^\theta(x, y_j))} \right] \\ &\leq \log N - H^{\mathbf{pop}_Y}(I|XY^N) \\ &\leq \min \{I(X, Y), \log N\} \end{aligned}$$

Thus minimizing $\bar{H}_\theta^{\mathbf{pop}_Y}(I|XY^N)$ over θ corresponds to maximizing a parameterized lower bound $B(\theta)$ on $I(X, Y)$, and for this reason global NCE is sometimes called “InfoNCE”. This lower bound cannot be greater than $\log N$, which is consistent with the result in [6].

Let $\theta^{q_Y} \in \arg \min_{\theta} \bar{H}_{\theta}^{q_Y}(I|XY^N)$. By universality we must have $p_{I|XY^N}^{\theta^{q_Y}} = \mathbf{pop}_{I|XY^N}^{q_Y}$. By (1) this means

$$s^{\theta^{q_Y}}(x, y) = \log \frac{\mathbf{pop}_{Y|X}(y|x)}{q_Y(y)} + \log C_x \quad \forall x \in \mathcal{X}, y \in \mathcal{Y}$$

for some constant $C_x > 0$. In particular, we can use the optimal parameter θ^{q_Y} to recover the underlying conditional distribution

$$\mathbf{pop}_{Y|X}(y|x) = \frac{\exp(s^{\theta^{q_Y}}(x, y) + \log q_Y(y))}{\sum_{y'} \exp(s^{\theta^{q_Y}}(x, y') + \log q_Y(y'))} \quad (2)$$

This is consistent with the “ranking” algorithm in [5]. Note that the additive adjustment is unnecessary if we choose uniformly random q_Y . A small modification of global NCE gives an unbiased gradient estimator of the cross entropy loss [1, 2] (Appendix C).

2 Local NCE

2.1 Model

The local NCE objective assumes a biased coin with head probability $1/N$, which we define by $\mathbf{pop}_A(1) = 1/N$ and $\mathbf{pop}_A(0) = (N-1)/N$. Given $x \sim \mathbf{pop}_X$ and $a \sim \mathbf{pop}_A$, it defines

$$\mathbf{pop}_{Y|XA}^{q_Y}(y|x, a) := \begin{cases} \mathbf{pop}_{Y|X}(y|x) & \text{if } a = 1 \\ q_Y(y) & \text{if } a = 0 \end{cases}$$

This yields the conditional head probability

$$\mathbf{pop}_{A|XY}^{q_Y}(1|x, y) = \frac{\mathbf{pop}_{Y|X}(y|x)}{\mathbf{pop}_{Y|X}(y|x) + (N-1)q_Y(y)} \quad (3)$$

Given $x \sim \mathbf{pop}_X$ and N iid samples $a_i \sim \mathbf{pop}_A$ and $y_i \sim \mathbf{pop}_{Y|XA}^{q_Y}(\cdot|x, a_i)$ for $i = 1 \dots N$, the joint conditional probability of the coin flips is given by

$$\mathbf{pop}_{A^N|XY^N}^{q_Y}(a_1 \dots a_N | x, y_1 \dots y_N) = \prod_{i=1: a_i=1}^N \mathbf{pop}_{A|XY}^{q_Y}(1|x, y_i) \prod_{j=1: a_j=0}^N (1 - \mathbf{pop}_{A|XY}^{q_Y}(1|x, y_j))$$

Let $H^{q_Y}(A^N|XY^N)$ denote the conditional entropy of $\mathbf{pop}_{A^N|XY^N}^{q_Y}$. We write it in the friendlier form (see Appendix D for details)

$$\begin{aligned} H^{q_Y}(A^N|XY^N) &= \mathbf{E}_{(x,y) \sim \mathbf{pop}_{XY}} \left[-\log \mathbf{pop}_{A|XY}^{q_Y}(1|x, y) \right] \\ &\quad + (N-1) \mathbf{E}_{\substack{x \sim \mathbf{pop}_X \\ y \sim q_Y}} \left[-\log(1 - \mathbf{pop}_{A|XY}^{q_Y}(1|x, y)) \right] \end{aligned} \quad (4)$$

The following lemma can be easily shown by plugging in (3) into (4) (again see Appendix D for details).

Lemma 2.1. Let $\text{KL}(p||q)$ denote the KL divergence between distributions p and q . Then

$$\begin{aligned} -H^{q_Y}(A^N|XY^N) &= \text{KL}\left(\mathbf{pop}_{Y|X} \left\| \frac{\mathbf{pop}_{Y|X} + (N-1)q_Y}{N}\right.\right) + (N-1)\text{KL}\left(q_Y \left\| \frac{\mathbf{pop}_{Y|X} + (N-1)q_Y}{N}\right.\right) \\ &\quad - \log N - (N-1)\log\left(\frac{N}{N-1}\right) \end{aligned}$$

Corollary 2.2. Let $\text{JSD}(p||q) = \frac{1}{2}\text{KL}(p||\frac{p+q}{2}) + \frac{1}{2}\text{KL}(q||\frac{p+q}{2})$ denote the Jensen-Shannon divergence. With $N = 2$ we have from Lemma 2.1

$$-H^{q_Y}(A^2|XY^2) = 2\text{JSD}\left(\mathbf{pop}_{Y|X} \left\| q_Y\right.\right) - \log 4$$

To make the connection to GANs [3] clear, let $|\mathcal{X}| = 1$ and eliminate the dependence on X . Recall the adversarial objective of GANs and its equilibrium:

$$\begin{aligned} \mathbf{GAN}(D, q_Y) &:= \mathbf{E}_{y \sim \mathbf{pop}_Y} [\log D(1|y)] + \mathbf{E}_{y \sim q_Y} [\log(1 - D(1|y))] \\ J_{\text{GAN}} &:= \min_{q_Y} \max_D \mathbf{GAN}(D, q_Y) \end{aligned}$$

where $D : \mathcal{Y} \rightarrow [0, 1]$ is a discriminator and q_Y is viewed as a generator. It can be verified that setting $D(1|y) = \mathbf{pop}_{A|Y}^{q_Y}(1|y) = \mathbf{pop}_Y(y)/(\mathbf{pop}_Y(y) + q_Y(y))$ (3) maximizes $\mathbf{GAN}(D, q_Y)$ for any q_Y . But $\mathbf{GAN}(\mathbf{pop}_{A|Y}^{q_Y}, q_Y) = -H^{q_Y}(A^2|Y^2)$, thus by Corollary 2.2

$$J_{\text{GAN}} = \min_{q_Y} \mathbf{GAN}(\mathbf{pop}_{A|Y}^{q_Y}, q_Y) = \min_{q_Y} 2\text{JSD}\left(\mathbf{pop}_Y \left\| q_Y\right.\right) - \log 4 = -\log 4$$

where the minimizer is $q_Y = \mathbf{pop}_Y$. At this equilibrium, we see that the best discriminator is uniform $\mathbf{pop}_{A|Y}^{\mathbf{pop}_Y}(1|y) = 1/2$ and the generator “wins”.

2.2 Estimation

We use a score function $s^\theta(x, y)$ through the sigmoid function to estimate $\mathbf{pop}_{A|XY}^{q_Y}$

$$p_{A|XY}^\theta(1|x, y) := \frac{1}{1 + \exp(-s^\theta(x, y))}$$

This is used to define the joint conditional distribution

$$p_{A^N|XY^N}^\theta(a_1 \dots a_N | x, y_1 \dots y_N) = \prod_{i=1: a_i=1}^N p_{A|XY}^\theta(1|x, y_i) \prod_{j=1: a_j=0}^N (1 - p_{A|XY}^\theta(1|x, y_j))$$

The model is again estimated by minimizing the cross (conditional) entropy between $\mathbf{pop}_{A^N|XY^N}^{q_Y}$ and $p_{A^N|XY^N}^\theta$. Similar to (4) this objective can be written in the friendlier form

$$\theta^{q_Y} \in \arg \max_{\theta} \mathbf{E}_{(x, y) \sim \mathbf{pop}_{XY}} [\log p_{A|XY}^\theta(1|x, y)] + (N-1) \mathbf{E}_{\substack{x \sim \mathbf{pop}_X \\ y \sim q_Y}} [\log(1 - p_{A|XY}^\theta(1|x, y))]$$

By universality we must have $p_{A|XY}^{\theta^{q_Y}} = \mathbf{pop}_{A|XY}^{q_Y}$. By (3) this means

$$s^{\theta^{q_Y}}(x, y) = \log \frac{\mathbf{pop}_{Y|X}(y|x)}{q_Y(y)} - \log(N-1) \quad \forall x \in \mathcal{X}, y \in \mathcal{Y}$$

If $q_Y = \mathbf{pop}_Y$, the optimal score of (x, y) is the pointwise mutual information (PMI) minus the log of the number of negative examples: this gives the analysis of the skip-gram objective of word2vec in [4]. We can use the optimal parameter θ^{q_Y} to recover the underlying conditional distribution

$$\mathbf{pop}_{Y|X}(y|x) = \exp \left(s^{\theta^{q_Y}}(x, y) + \log q_Y(y) + \log(N-1) \right)$$

This is consistent with the “binary” algorithm in [5]. Note that unlike (2) this calculation doesn’t require normalization. This implies that the score function must self-normalized (Assumption 2.2 in [5]), that is we must be able to at least find θ such that

$$\sum_y \exp \left(s^\theta(x, y) + \log q_Y(y) + \log(N-1) \right) = 1 \quad \forall x \in \mathcal{X}$$

This is a strong assumption when $|\mathcal{X}|$ is larger than the number of variables in θ , so universality cannot be taken for granted in this case.

A Hinge Loss

We want to find θ that maximizes the probability of the event that $s^\theta(x, y) > s^\theta(x, y')$ for all $y' \neq y$. This is equivalent to minimizing the **zero-one loss**

$$\arg \min_{\theta} \mathbf{E}_{(x,y) \sim \mathbf{pop}_{XY}} \left[\mathbb{1} \left(\overbrace{s^\theta(x, y) - \max_{y' \neq y} s^\theta(x, y')}^{\text{zero-one loss on } (x, y)} \leq 0 \right) \right]$$

margin of (x, y)

where $\mathbb{1}(\cdot) \in \{0, 1\}$ is the indicator function. The indicator function is difficult to optimize for a number of reasons (e.g., it has zero gradient almost everywhere wrt the margin), so we instead define the **hinge loss**

$$\arg \min_{\theta} \mathbf{E}_{(x,y) \sim \mathbf{pop}_{XY}} \left[\max \left\{ 0, 1 - \underbrace{\left(s^\theta(x, y) - \max_{y' \neq y} s^\theta(x, y') \right)}_{\text{margin of } (x, y)} \right\} \right]$$

hinge loss on (x, y)

Note that for any fixed (x, y) , the hinge loss on (x, y) is a convex upper bound on the zero-one loss on (x, y) where the convexity is wrt the margin of (x, y) .

In some applications, it's neither necessary nor useful to exactly maximize over the negative space $\{y' \in \mathcal{Y} : y' \neq y\}$ to compute the margin. This is because the search is intractable and/or exact maximization has some undesirable quality (e.g., it's in fact an alternative viable prediction). In this case, maximization is replaced by sampling [11].

B Cross-Entropy Loss

We frame the problem as conditional density estimation of $\mathbf{pop}_{Y|X}$. To this end, we turn the score function into a proper conditional distribution by using the softmax operation:

$$p_{Y|X}^\theta(y|x) := \frac{\exp(s^\theta(x, y))}{\sum_{y'} \exp(s^\theta(x, y'))} \quad \forall x \in \mathcal{X}, y \in \mathcal{Y}$$

Then we find θ that minimizes the cross (conditional) entropy between $\mathbf{pop}_{Y|X}$ and $p_{Y|X}^\theta$:

$$\theta^* \in \arg \min_{\theta} \mathbf{E}_{(x,y) \sim \mathbf{pop}_{XY}} \left[-\log p_{Y|X}^\theta(y|x) \right] \quad (5)$$

By universality we must have $p_{Y|X}^{\theta^*} = \mathbf{pop}_{Y|X}$. This means

$$\frac{\exp(s^{\theta^*}(x, y))}{\sum_{y'} \exp(s^{\theta^*}(x, y'))} = \frac{\mathbf{pop}_{XY}(x, y)}{\sum_{y'} \mathbf{pop}_{XY}(x, y')} \quad \forall x \in \mathcal{X}, y \in \mathcal{Y}$$

and it follows that $\exp(s^{\theta^*}(x, y)) = C_x \mathbf{pop}_{XY}(x, y)$ for some $C_x > 0$. Hence

$$s^{\theta^*}(x, y) = \log \mathbf{pop}_{XY}(x, y) + \log C_x \quad \forall x \in \mathcal{X}, y \in \mathcal{Y}$$

That is, the optimal score of (x, y) is the log probability of (x, y) shifted by some constant dependent on x .

C Gradient Estimation

Without loss of generality we consider the following simplified setting. Fix some target $t \in \mathcal{X}$ and define the loss function of $\theta \in \mathbb{R}^{|\mathcal{X}|}$ by

$$L(\theta) := -\log \frac{\exp(\theta_t)}{\sum_{x \in \mathcal{X}} \exp(\theta_x)} = \log Z(\theta) - \theta_t$$

where $Z(\theta) := \sum_{x \in \mathcal{X}} \exp(\theta_x)$. Now, let q be any full-support distribution over $\mathcal{X} \setminus \{t\}$. For any $\underline{n} = (n_1 \dots n_m) \in (\mathcal{X} \setminus \{t\})^m$ we define

$$\hat{L}_{q, \underline{n}}(\theta) := -\log \frac{\exp(\theta_t)}{\exp(\theta_t) + \frac{1}{m} \sum_{i=1}^m \frac{\exp(\theta_{n_i})}{q(n_i)}} = \log \hat{Z}_{q, \underline{n}}(\theta) - \theta_t$$

where $\hat{Z}_{q, \underline{n}}(\theta) := \exp(\theta_t) + \frac{1}{m} \sum_{i=1}^m \frac{\exp(\theta_{n_i})}{q(n_i)}$.

Lemma C.1.

$$\mathbf{E}_{\underline{n} \sim q^m} [\hat{Z}_{q, \underline{n}}(\theta)] = Z(\theta)$$

Proof.

$$\begin{aligned} \mathbf{E}_{\underline{n} \sim q^m} [\hat{Z}_{q, \underline{n}}(\theta)] &= \exp(\theta_t) + \mathbf{E}_{\underline{n} \sim q^m} \left[\frac{1}{m} \sum_{i=1}^m \frac{\exp(\theta_{n_i})}{q(n_i)} \right] \\ &= \exp(\theta_t) + \mathbf{E}_{n \sim q} \left[\frac{\exp(\theta_n)}{q(n)} \right] \\ &= \exp(\theta_t) + \sum_{n \in \mathcal{X} \setminus \{t\}} q(n) \frac{\exp(\theta_n)}{q(n)} \\ &= \sum_{x \in \mathcal{X}} \exp(\theta_x) \\ &= Z(\theta) \end{aligned}$$

□

It is convenient to define $\phi_{q,\underline{n}}(\theta) \in \mathbb{R}^{m+1}$ where

$$[\phi_{q,\underline{n}}(\theta)]_i = \begin{cases} \theta_{n_i} - \log(mq(n_i)) & \text{if } i < m+1 \\ \theta_t & \text{otherwise} \end{cases}$$

We can now write $\widehat{L}_{q,\underline{n}}(\theta) = -\log p_{\phi_{q,\underline{n}}(\theta)}(m+1)$ where

$$p_{\phi_{q,\underline{n}}(\theta)}(i) := \frac{\exp([\phi_{q,\underline{n}}(\theta)]_i)}{\sum_{j=1}^{m+1} \exp([\phi_{q,\underline{n}}(\theta)]_j)} \quad \forall i \in \{1 \dots m+1\}$$

Let $p_\theta(x) := \exp(\theta_x) / \sum_{x' \in \mathcal{X}} \exp(\theta_{x'})$ denote the full softmax. The following gradient expressions are easy to verify:

$$\nabla L(\theta) = \mathbf{E}_{x \sim p_\theta} [\mathbb{1}_x] - \mathbb{1}_t \quad (6)$$

$$\nabla_{\underline{n} \sim q^m} \mathbf{E} [\widehat{L}_{q,\underline{n}}(\theta)] = \mathbf{E}_{\substack{\underline{n} \sim q^n \\ i \sim p_{\phi_{q,\underline{n}}(\theta)}}} [\nabla[\phi_{q,\underline{n}}(\theta)]_i] - \mathbb{1}_t \quad (7)$$

where $\mathbb{1}_x \in \{0,1\}^{|\mathcal{X}|}$ denotes a one-hot vector with 1 at index x .

Lemma C.2. $\nabla L(\theta) = \nabla_{\underline{n} \sim q^m} \mathbf{E} [\widehat{L}_{q,\underline{n}}(\theta)]$ iff $q(x) \propto \exp(\theta_x)$ for all $x \in \mathcal{X}$.

Proof. From (6) and (7) it is clear that the statement is equivalent to

$$p_\theta(l) = \mathbf{E}_{\substack{\underline{n} \sim q^n \\ i \sim p_{\phi_{q,\underline{n}}(\theta)}}} \left[\frac{\partial[\phi_{q,\underline{n}}(\theta)]_i}{\partial \theta_l} \right] = \mathbf{E}_{\underline{n} \sim q^n} \left[\sum_{i=1}^{m+1} \frac{\exp([\phi_{q,\underline{n}}(\theta)]_i)}{\widehat{Z}_{q,\underline{n}}(\theta)} \frac{\partial[\phi_{q,\underline{n}}(\theta)]_i}{\partial \theta_l} \right] \quad (8)$$

for all $l \in \mathcal{X}$, iff $q(x) \propto \exp(\theta_x)$ for all $x \in \mathcal{X}$.

- $l = t$: In this case we have

$$\frac{\partial[\phi_{q,\underline{n}}(\theta)]_i}{\partial \theta_t} = \begin{cases} 1 & \text{if } i = m+1 \\ 0 & \text{otherwise} \end{cases}$$

Therefore the last term of (8) is

$$\mathbf{E}_{\underline{n} \sim q^n} \left[\frac{\exp(\theta_t)}{\widehat{Z}_{q,\underline{n}}(\theta)} \right] = \frac{\exp(\theta_t)}{\mathbf{E}_{\underline{n} \sim q^n} [\widehat{Z}_{q,\underline{n}}(\theta)]} = \frac{\exp(\theta_t)}{Z(\theta)} = p_\theta(t)$$

Note that this holds for any choice of q .

- $l \neq t$: In this case we have

$$\frac{\partial[\phi_{q,\underline{n}}(\theta)]_i}{\partial \theta_l} = \begin{cases} [[n_i = l]] & \text{if } i < m+1 \\ 0 & \text{otherwise} \end{cases}$$

Therefore the last term of (8) is

$$\mathbf{E}_{\underline{n} \sim q^n} \left[\frac{1}{\widehat{Z}_{q,\underline{n}}(\theta)} \sum_{i=1}^m \frac{\exp(\theta_{n_i})}{mq(n_i)} [[n_i = l]] \right] \stackrel{*}{=} \frac{\mathbf{E}_{\underline{n} \sim q} \left[\frac{\exp(\theta_n)}{q(n)} [[n = l]] \right]}{\mathbf{E}_{\underline{n} \sim q^n} [\widehat{Z}_{q,\underline{n}}(\theta)]} = \frac{\exp(\theta_l)}{Z(\theta)} = p_\theta(l)$$

where the equality with $*$ holds iff $\widehat{Z}_{q,\underline{n}}(\theta) = \exp(\theta_t) + \frac{1}{m} \sum_{i=1}^m \frac{\exp(\theta_{n_i})}{q(n_i)}$ is constant for all $\underline{n} \in (\mathcal{X} \setminus \{t\})^m$. This implies that $q(x) \propto \exp(\theta_x)$ for all $x \in \mathcal{X}$.

□

Define a distribution q_θ^* over $\mathcal{X} \setminus \{t\}$ by

$$q_\theta^*(n) = \frac{\exp(\theta_n)}{\sum_{x \in \mathcal{X} \setminus \{t\}} \exp(\theta_x)}$$

We see that indeed for any $\underline{n} \in (\mathcal{X} \setminus \{t\})^m$,

$$\widehat{L}_{q_\theta^*, \underline{n}}(\theta) = -\log \frac{\exp(\theta_t)}{\exp(\theta_t) + \frac{1}{m} \sum_{i=1}^m \frac{\exp(\theta_{n_i})}{q_\theta^*(n_i)}} = -\log \frac{\exp(\theta_t)}{\exp(\theta_t) + \sum_{x \in \mathcal{X} \setminus \{t\}} \exp(\theta_x)} = L(\theta)$$

Getting q_θ^* requires computing a normalization term $\sum_{x \in \mathcal{X} \setminus \{t\}} \exp(\theta_x)$ for each target $t \in \mathcal{X}$. As a more efficient alternative in practice, we can approximate this distribution by p_θ and exclude sampled targets. The bias of the gradient estimator using an approximate $\hat{q}_\theta \neq q_\theta^*$ is analyzed in [10].

D Detailed Derivations

To get (4), note that

$$\begin{aligned} & -H^{q_Y}(A^N | XY^N) \\ &= \mathbf{E}_{\substack{x \sim \mathbf{pop}_X \\ a_i \sim \mathbf{pop}_A, y_i \sim \mathbf{pop}_{Y|XA}^{q_Y}(\cdot|x, a_i)}} \left[\log \mathbf{pop}_{A^N|XY^N}^{q_Y}(a_1 \dots a_N | x, y_1 \dots y_N) \right] \\ &= \mathbf{E}_{\substack{x \sim \mathbf{pop}_X \\ a_i \sim \mathbf{pop}_A, y_i \sim \mathbf{pop}_{Y|XA}^{q_Y}(\cdot|x, a_i)}} \left[\sum_{i=1}^N [[a_i = 1]] \log \mathbf{pop}_{A|XY}^{q_Y}(1|x, y_i) + [[a_i = 0]] \log(1 - \mathbf{pop}_{A|XY}^{q_Y}(1|x, y_i)) \right] \\ &= N \mathbf{E}_{\substack{x \sim \mathbf{pop}_X \\ a \sim \mathbf{pop}_A, y \sim \mathbf{pop}_{Y|XA}^{q_Y}(\cdot|x, a)}} \left[[[a = 1]] \log \mathbf{pop}_{A|XY}^{q_Y}(1|x, y) + [[a = 0]] \log(1 - \mathbf{pop}_{A|XY}^{q_Y}(1|x, y)) \right] \end{aligned}$$

Use the tower rule $\mathbf{E}[X] = \mathbf{E}[\mathbf{E}[X|Y]]$ on each term of the expectation. For the first term,

$$\begin{aligned} N \mathbf{E}_{\substack{x \sim \mathbf{pop}_X \\ a \sim \mathbf{pop}_A, y \sim \mathbf{pop}_{Y|XA}^{q_Y}(\cdot|x, a)}} \left[[[a = 1]] \log \mathbf{pop}_{A|XY}^{q_Y}(1|x, y) \right] &= N \left(\frac{1}{N} \mathbf{E}_{\substack{x \sim \mathbf{pop}_X \\ y \sim \mathbf{pop}_{Y|X}(\cdot|x)}} \left[\log \mathbf{pop}_{A|XY}^{q_Y}(1|x, y) \right] \right) \\ &= \mathbf{E}_{(x,y) \sim \mathbf{pop}_{XY}} \left[\log \mathbf{pop}_{A|XY}^{q_Y}(1|x, y) \right] \end{aligned}$$

For the second term,

$$\begin{aligned} N \mathbf{E}_{\substack{x \sim \mathbf{pop}_X \\ a \sim \mathbf{pop}_A, y \sim \mathbf{pop}_{Y|XA}^{q_Y}(\cdot|x, a)}} \left[[[a = 0]] \log(1 - \mathbf{pop}_{A|XY}^{q_Y}(1|x, y)) \right] &= N \left(\frac{N-1}{N} \mathbf{E}_{\substack{x \sim \mathbf{pop}_X \\ y \sim q_Y}} \left[\log(1 - \mathbf{pop}_{A|XY}^{q_Y}(1|x, y)) \right] \right) \\ &= (N-1) \mathbf{E}_{\substack{x \sim \mathbf{pop}_X \\ y \sim q_Y}} \left[\log(1 - \mathbf{pop}_{A|XY}^{q_Y}(1|x, y)) \right] \end{aligned}$$

To get Lemma 2.1, first note that $(\mathbf{pop}_{Y|X}(\cdot|x) + (N-1)q_Y)/N$ is a proper conditional distribution over $\mathcal{Y}$. The first term of $-H^{q_Y}(A^N|XY^N)$ is

$$\begin{aligned} \mathbf{E}_{(x,y) \sim \mathbf{pop}_{XY}} \left[\log \mathbf{pop}_{A|XY}^{q_Y}(1|x, y) \right] &= \mathbf{E}_{(x,y) \sim \mathbf{pop}_{XY}} \left[\log \frac{\mathbf{pop}_{Y|X}(y|x)}{\mathbf{pop}_{Y|X}(y|x) + (N-1)q_Y(y)} \right] \\ &= \mathbf{E}_{(x,y) \sim \mathbf{pop}_{XY}} \left[\log \frac{\frac{\mathbf{pop}_{Y|X}(y|x)}{N}}{\frac{\mathbf{pop}_{Y|X}(y|x) + (N-1)q_Y(y)}{N}} \right] \\ &= \text{KL} \left(\mathbf{pop}_{Y|X} \left\| \frac{\mathbf{pop}_{Y|X} + (N-1)q_Y}{N} \right\| \right) - \log N \end{aligned}$$

The second term of $-H^{q_Y}(A^N|XY^N)$ is similarly

$$\begin{aligned} (N-1) \mathbf{E}_{\substack{x \sim \mathbf{pop}_X \\ y \sim q_Y}} \left[\log(1 - \mathbf{pop}_{A|XY}^{q_Y}(1|x, y)) \right] &= (N-1) \mathbf{E}_{\substack{x \sim \mathbf{pop}_X \\ y \sim q_Y}} \left[\log \frac{(N-1)q_Y(y)}{\mathbf{pop}_{Y|X}(y|x) + (N-1)q_Y(y)} \right] \\ &= (N-1) \text{KL} \left(q_Y \left\| \frac{\mathbf{pop}_{Y|X} + (N-1)q_Y}{N} \right\| \right) - (N-1) \log \frac{N}{N-1} \end{aligned}$$

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